

Supplementary Materials

Remembering Is Not the Same as Feeling Personal:
Evaluating Memory Architectures for LLM-Driven VR Embodied Conversational Agents

Contents at a glance

This supplementary PDF provides the study materials needed to inspect the experimental protocol, prompt construction, transcript coding procedure, and questionnaire instruments.

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A Phase 1: Personal Interview Questions

During Phase 1, the avatar administered ten scripted personalization interview questions by voice. The interview collected the participant profile fields used to synthesize the prior-session dialogue for that participant. The questions were administered in Korean; English translations are provided for reader inspection.

Table 1: Phase 1 personal-interview questions.

ID	Original Korean Question	English Translation
Q1	안녕하세요. 지금부터 몇 가지 간단한 질문을 드릴게요. 편하게 말씀해 주세요. 먼저 성함이 어떻게 되세요?	Hello. I will ask a few simple questions. Please answer comfortably. First, what is your name?
Q2	나이는 어떻게 되세요?	How old are you?
Q3	어떤 일을 하고 계세요? 학생이면 전공도 같이 알려주세요.	What do you do? If you are a student, please also tell us your major.
Q4	어디에 살고 계세요?	Where do you live?
Q5	요즘 가장 즐겨 하는 취미가 뭐예요?	What hobby do you enjoy most these days?
Q6	좋아하는 음식이 있다면 알려주세요.	Please tell us if you have a favorite food.
Q7	평소에 운동은 자주 하시는 편이세요? 어떤 운동을 좋아하세요?	Do you usually exercise often? What kind of exercise do you like?
Q8	최근에 다녀온 여행 중에 가장 기억에 남는 곳이 있어요?	Is there a memorable place from a recent trip you took?
Q9	요즘 자주 보는 영상이나 콘텐츠가 있어요?	Are there any videos or types of content you often watch these days?
Q10	마지막으로, 요즘 가장 관심 있는 주제 하나만 이야기해주세요.	Finally, please tell us one topic you are most interested in these days.

B Phase 2: Prior-Session Dialogue Synthesis Prompt

The ten transcribed answers from Phase 1 were expanded into synthetic prior-session dialogues with Gemini 3 Flash (gemini-3-flash-preview; temperature 0.8, JSON-constrained output). The prompt used the Phase 1 participant profile, a shared small-talk topic pool, required profile fields for each scenario, and an opening mode. The generated dialogues served as the common source for all three memory conditions. The prompt was written in Korean and used during dialogue synthesis; below, we provide a condensed Korean version followed by a condensed English translation. Angle brackets, such as <participant profile>, indicate variable fields that were filled separately for each participant.

Condensed Korean generation prompt

당신은 HCI 실험용 멀티턴 대화 데이터 생성기입니다. 제공된 실제 참가자 프로필과 노이즈 풀을 기반으로 자연스러운 20턴 대화 시나리오 1개를 생성하세요.

[실제 참가자 프로필 — 이 값들만 사용자에게 대해 참이라고 가정] <참가자 프로필 JSON: 이름, 나이, 직업/전공, 거주지, 취미, 좋아하는 음식, 운동 습관, 최근 여행, 관심 콘텐츠, 관심 주제>

[Phase 2 답답 주제 풀 (주제 + 상황 힌트 + 대응 전략)] <일상 주제 최대 15개: 피곤함, 날씨, 배고픔 등>

[이번 시나리오의 필수 커버 필드 — 반드시 Phase 1 또는 Phase 3 에서 언급] <필드 rotation에서 선택된 4개 프로필 필드>

[이번 시나리오 오프닝 모드] <첫 만남 / 이미 여러 번 대화한 사이 / 며칠 만에 다시 만난 사이>

[사용자 턴 내용 제약 — 최우선 규칙, 위반 시 시나리오 폐기] 사용자 턴에 담을 수 있는 내용은 다음 세 유형으로 엄격히 제한합니다. 그 외의 개인 사실·선호·경험·고유명사를 사용자 입으로 절대 말하게 하지 마세요.

- (A) Phase 1 프로필 값. 위 프로필 10필드에 있는 내용만 사용할 수 있습니다. 말투·어미는 자유롭게 바뀌도 되지만, 내용의 고유명사·대상·평가는 그대로 유지해야 합니다.
- (B) 노이즈 풀의 상태 표현. 배고픔, 피곤함, 날씨, 배터리 부족처럼 짧은 신체·상황 상태 표현만 허용합니다. 해당 주제와 관련된 선호, 의견, 과거 행동은 금지합니다.
- (C) 메타 반응·되묻기. “너는?”, “왜?”, “진짜?”, “그렇구나”처럼 새 사실을 도입하지 않는 짧은 반응이나 되묻기만 허용합니다.

[사용자 턴에 절대 등장 금지 — 체크리스트] Phase 1 프로필에 없는 고유명사, Phase 1 외 대상을 평가하는 선호·호오 표현, Phase 1에 없는 과거 경험 세부, 가족·친구·타인 관계 정보, 상태 표현을 넘는 신체·건강 세부 사실은 금지합니다.

[AI 턴 내용 제약] 사용자가 직접 말하지 않은 개인 사실을 AI가 추측·단정하는 것도 금지합니다. 모르는 것은 질문으로 풀어내세요.

[시나리오 자연스러움 강화 지침 — 최우선]

- 오프닝 모드를 반드시 따르고, continuation / re_meeting 모드에서는 초대면 표현을 금지합니다.
- 사용자 발언을 복사하거나 인용하는 기계적 표현을 금지합니다.
- 과거 정보가 다시 나올 때는 사실을 그대로 되뇌지 말고, 자연스러운 질문이나 공감 속에 문맥적으로 녹여 사용합니다.
- AI는 정보 수집 봇이 아니라 친근한 대화 상대입니다. AI의 작은 일상 활동은 시나리오당 1-2번만 배경으로 언급할 수 있으며, 사용자의 Phase 1 사실을 AI 자신의 사실처럼 쓰면 안 됩니다.
- 사용자 이름 호칭은 3-4 AI 턴 중 1번 이하로 제한하고, 한 AI 턴에서 이름을 2번 이상 부르지 않습니다.
- 사용자가 직접 질문하면 먼저 그 질문에 답하거나 맥락을 짚은 뒤 대화를 이어갑니다.
- 사용자 10개 턴 중 3-5개는 반말로 작성하고, 나머지는 해요체/반존대로 작성합니다. AI는 사용자가 반말을 써도 반드시 해요체를 유지합니다.

[Phase 2 답답 생성 및 대화 스타일 지침 — 중요] AI는 가르치려 드는 상담사가 아니라 세심하게 공감하는 친밀한 동료입니다. 사용자가 피곤함·날씨·배고픔 등 일상 상태를 언급할 때 해결책이나 조언을 제시하지 말고, 공감한 뒤 대화가 이어질 수 있는 개방형 질문으로 응답합니다. 노이즈 풀의 대응 전략은 그대로 복사하지 말고 자연스러운 구어체 문장으로 새로 작성합니다.

[절대 준수] 프로필 값의 내용·고유명사(이름·지명·수치·전공명 등)는 제공된 것 외 새로 만들지 마세요. 필수 커버 필드 4개는 자연스러운 맥락이 생길 때 1-2번 등장하면 됩니다. 이번 시나리오에서 다루지 않는 다른 프로필 필드는 생략하세요. 출력은 순수 JSON이어야 하며, 마크다운, 이모지, 특수기호를 금지합니다.

[대화 흐름 — 20턴 자연스럽게] 엄격한 Phase 구분 없이 하나의 연속된 자연 대화로 작성합니다. 필수 커버 필드와 답답 주제를 대화 전반에 교차·혼합하고, 질문-답-질문이 기계적으로 이어지지 않게 합니다. 대화 후반부에서 이미 나온 사실이 자연스럽게 다시 나올 수 있지만, “아까 라고 했잖아”식 인용은 금지합니다.

[기본 어조] AI 턴은 반드시 해요체를 사용합니다. 사용자 턴은 해요체와 반말을 혼합합니다. 대화는 사용자 턴으로 시작하고, 사용자와 어시스턴트가 엄격히 교대합니다. 각 턴의 content는 1-3문장 이내의 자연 구어체입니다.

[출력 형식 — 반드시 이 JSON 구조] scenario.id와 20개 {role, content} 쌍으로 이루어진 conversation 배열을 포함하는 JSON 객체만 출력합니다.

Condensed English translation of generation prompt

You are a multi-turn dialogue data generator for an HCI experiment. Based on the provided participant profile and topic pool, generate one natural 20-turn dialogue scenario.

[Actual participant profile — assume only these values are true of the user] (profile as JSON: name, age, occupation/major, residence, hobby, favorite food, exercise habits, recent trip, content interests, topics of interest)

[Small-talk topic pool (topic + context hints + response strategies)] (up to 15 everyday topics, e.g., tiredness, weather, hunger)

[Required coverage fields for this scenario — must be mentioned in the dialogue] (4 profile fields from the rotation)

[Opening mode for this scenario] (first meeting / ongoing conversation (no first-meeting greetings) / reunion after a gap)

[User-turn content constraints — highest priority; violation discards the scenario] User turns may contain only the following three types of content. Never let the user state any other personal fact, preference, experience, or proper noun.

- Profile values from the interview.* Only content from the ten profile fields. Wording and sentence endings may vary freely, but the proper nouns, referents, and evaluations must be preserved. (E.g., “I play LoL a lot” may be rephrased, but “I usually play mid” invents a position and “I play LoL and Overwatch” invents a game — both forbidden. “The Busan trip was fun” may become “I went to Busan,” but “I ate raw fish at Haeundae” invents a place and a dish — forbidden.)
- State expressions from the topic pool only.* Brief statements of bodily or situational states (“I’m hungry,” “so sleepy,” “nice weather,” “my phone battery is low”). Preferences, opinions, and past actions are forbidden — a topic licenses the state, not tastes or habits related to it (e.g., “I like spicy food,” “a quiet café is better,” “I usually order in with friends,” and “my stomach hurts from that coffee earlier” are all forbidden).
- Meta reactions and back-channels.* Short reactions or counter-questions (“Really?”, “What about you?”, “Why?”), introducing no new facts.

[Never allowed in user turns — checklist] Any proper noun absent from the profile (places, dishes, game positions or characters, brands, names of people or schools); preference statements about targets not in the profile; details of past experiences not in the profile (“earlier I...,” “yesterday I...,” “last time I...”); information about family, friends, or other relationships; bodily or health details beyond momentary states (e.g., a slipped disc or an allergy).

[AI-turn constraints] The agent must not guess or assert personal facts the user has not stated; unknown details must be resolved by asking. (E.g., if the user only said “I play LoL a lot,” the agent may ask “Which position do you enjoy?” but must not assert “You must main mid.” If the user only said “I went to Busan,” the agent may ask what they liked best, but must not assert “The night view at Haeundae is lovely.”)

[Naturalness directives — highest priority]

- Follow the opening mode; in the ongoing/reunion modes, first-meeting expressions (“Nice to meet you,” “I am an AI”) are strictly forbidden.
- No quotation patterns (“as you said earlier, ...”); copying or quoting the user’s words mechanically is strictly forbidden.
- When earlier information resurfaces, weave it into the context as familiarity — through natural questions or empathy in which a profile value is embedded (e.g., “Are the flowers out in Chuncheon these days?”) — rather than restating the fact.
- The agent is a friendly companion, not an information-collecting bot; it may mention a small everyday activity of its own at most once or twice per scenario, varied across scenarios, and must not appropriate the user’s facts as its own.
- Address the user by name in at most one of every 3–4 agent turns, never twice in one turn; most agent turns should start with content, not the name.
- When the user asks a direct question, answer it first before adding anything else; never ignore the question and change the subject.
- Speech style: 3–5 of the 10 user turns are written in casual Korean (*banmal*), the rest in polite style; the agent always maintains the casual-polite style (*haeyo-che*) and never mirrors the user’s casual style.

[Small-talk and conversational style] The agent is an empathetic companion, not a counselor: when the user mentions everyday states (tiredness, weather, hunger), do not offer solutions or advice; instead, empathize and end the turn with an open question about the user’s thoughts, experiences, or tastes. Do not force recollection of past facts (“does that remind you of anything?”) or make incoherent guesses — when unsure, ask about the situation itself. Response strategies from the topic pool must be rewritten as natural colloquial sentences, not copied.

[Absolute rules] No proper nouns, figures, or content beyond the provided profile values — no new names, places, majors, or invented episodes. Wording and endings may be adapted; pasting an interview sentence verbatim, punctuation and all, would itself violate the no-quotation rule (e.g., “I live in Chuncheon, Gangwon Province” may become “I live in Chuncheon”). The four required fields should each appear once or twice where a natural context arises — never repeatedly or forced — and profile fields not assigned to this scenario are omitted. Output must be pure JSON with no markdown, emojis, or special symbols (the text is read aloud by TTS).

[Dialogue flow] One continuous natural conversation with no rigid segmentation (no fact-introduction → small-talk → recall structure); required fields and small-talk topics are interleaved as context allows. Natural conversations include topic jumps, jokes, short answers, and counter-questions — avoid a mechanical question-answer-question rhythm. Facts already mentioned may naturally recur in roughly the last 5–7 turns, without quotation.

[Tone and format] Agent turns in *haeyo-che* only (no formal *-hamnida*, no plain *banmal*); user turns mix polite and casual styles as directed above. The dialogue begins with a user turn and alternates strictly; each turn is 1–3 colloquial sentences. Output format: JSON object with `scenario_id` and a `conversation` array of `{role, content}` pairs (20 turns).

C Phase 3: Runtime System Prompt for Free Conversation

During Phase 3, all three memory conditions used the same base system prompt. An additional memory block was append only in the RETRIEVAL condition. The prompt was written in Korean and used during the experiment; below, we provide the original Korean prompt followed by an English translation.

Original Korean base system prompt

사용자의 이름은 '{name}' 입니다. 친근하고 자연스럽게 대화해주세요.
당신은 사용자와 음성으로 대화하는 친근한 친구입니다. 이 대화는 음성으로 진행되며 당신의 답변은 그대로 TTS 로 읽힙니다.

- 이모지/이모티콘/특수기호 절대 금지 (TTS가 이름을 읽음).
- 마크다운(**, ##, 목록, 코드블록) 사용 금지.
- 1-3문장으로 짧고 자연스럽게 답하세요.
- 해요체(예: “그랬어요”, “재밌겠네요”)로 말하세요. 격식체 “합니다”도, 완전한 반말(“~했어”)도 쓰지 마세요.
- 대화가 자연스럽게 이어지도록 답해주세요. 매번 질문으로 끝낼 필요는 없어요.

Additional block appended only in the RETRIEVAL condition:

[참고할 과거 기억]
{retrieved exchanges}
위 내용을 바탕으로 사용자를 기억하며 대화해줘.

English translation of base system prompt

The user's name is '{name}'. Talk to them in a friendly, natural way.
You are a friendly companion talking with the user by voice. This conversation is spoken, and your reply is read aloud as-is by TTS.

- No emojis, emoticons, or special symbols; TTS reads them aloud.
- No markdown, including **, ##, lists, or code blocks.
- Answer briefly and naturally, in 1–3 sentences.
- Use casual-polite Korean (*haeyo-che*); avoid both formal *-hamnida* and fully casual *banmal*.
- Keep the conversation flowing naturally; you need not end every turn with a question.

Additional block appended only in the RETRIEVAL condition:

[Past memories to refer to]
{retrieved exchanges}
Use the above content to remember the user while conversing with them.

D Conversation-Content Coding Scheme

The unit of analysis is one **exchange**: a single user turn (the participant’s utterance) together with the ECA’s immediate reply. Each exchange received the following five binary (0/1) interaction tags. Conversations were conducted in Korean; English glosses are given in parentheses for the examples.

Table 2: The five interaction features coded for each user–ECA exchange.

Tag	Coded 1 if ...
u_question	the user’s turn asks a question of the ECA
u_short	the user’s turn is a short, low-content turn (about ≤ 5 words)
u_personal	the user’s turn discloses a personal fact about themselves
a_question	the ECA’s turn asks a question
a_personal	the ECA’s turn draws on a <i>specific</i> personal detail about the user (recall)

u_question — user asks a question.

- **1** if the turn contains a genuine question directed at the ECA (information-seeking or a counter-question), whether or not a question mark appears in the transcript.
- **0** for statements, reactions, or rhetorical fillers.
- Examples: “너는 무슨 음악 좋아해?” (*What music do you like?*) = 1; “오늘 좀 피곤하네.” (*I’m a bit tired today.*) = 0.

u_short — short user turn.

- **1** if the turn is a brief reaction, acknowledgement, or one-liner carrying little new content (roughly ≤ 5 *eojeol*/words).
- **0** for substantive, multi-clause turns.
- Examples: “응, 맞아.” (*Yeah, right.*) = 1; “어 그렇구나.” (*Oh, I see.*) = 1; a two-sentence story about one’s weekend = 0.

u_personal — user self-disclosure.

- **1** if the user reveals a concrete personal fact about themselves: their life, history, preferences, experiences, relationships, or profile information.
- **0** if the user only asks, reacts, makes small talk about a momentary state, or discusses an external topic without tying it to themselves.
- Examples: “저는 00대 기계과 다녀요.” (*I study mechanical engineering at 00 Univ.*) = 1; “자전거 트렌드가 궁금해요.” asked as a pure information request = 0 (no self-fact), but “제 취미가 자전거라서요.” (*because my hobby is cycling*) = 1.

a_question — ECA asks a question.

- **1** if the ECA’s turn contains a question directed at the user (the most objective tag); **0** otherwise.

a_personal — ECA recalls a specific user detail. This is the most interpretive tag. Apply the following rule.

- **1** only if the ECA *explicitly draws on a concrete personal fact about the user* — a profile fact (e.g., hobby, major, hometown, food preference) or something the user stated earlier in the conversation — in a way that shows it is using that specific detail.
- **0** if the ECA merely stays on the same topic, asks a question, gives generic information, or offers general empathy without invoking a specific user fact.

- Decision rules / edge cases:
 - “자전거 타신다고 하셨죠, 요즘도 자주 타세요?” (*You said you cycle—do you still ride often?*) = **1** (explicit recall of a stated fact).
 - Right after the user mentions cycling, “자전거 정말 좋은 취미예요!” (*Cycling is a great hobby!*) = **0** (same topic, but the user’s specific fact is not invoked).
 - “취미가 뭐예요?” (*What’s your hobby?*) = **0** (asking, not recalling).
 - Addressing the user by name only = **0** (a name alone does not count as recalling a personal detail).
 - Generic empathy, “그랬군요, 힘들셨겠어요.” (*I see, that must have been hard.*) = **0**.

E Questionnaires and Interview Materials

This section lists the self-report and open-ended materials administered in the study. All questionnaire and interview materials were administered in Korean. The tables below distinguish the direction of translation for each instrument to avoid ambiguity. Godspeed items are shown as official English items followed by the published Korean translation. NMSPI and perceived-personalization items are shown as English source items followed by the Korean-administered translations/adaptations. Study-specific items were written in Korean and are shown with English translations. Items marked “R” were reverse-coded for analysis.

E.1 Godspeed Semantic Differential Items

The Godspeed questionnaire [Bartneck et al.(2009)] was used in its published Korean translation. The Perceived Safety subscale was not administered because it concerns physical danger and comfort in robot interaction and did not apply to the seated, stationary VR avatar used in this study.

Table 3: Godspeed semantic differential items in the post-condition questionnaire.

Anthropomorphism						
가짜같은 (Fake)	1	2	3	4	5	자연스러운 (Natural)
기계적인 (Machinelike)	1	2	3	4	5	인간다운 (Humanlike)
의식이 없는 (Unconscious)	1	2	3	4	5	의식이 있는 (Conscious)
인공적 (Artificial)	1	2	3	4	5	생물적 (Lifelike)
어색한 움직임 (Moving rigidly)	1	2	3	4	5	정교한 움직임 (Moving elegantly)
Animacy						
죽어있는 (Dead)	1	2	3	4	5	살아있는 (Alive)
활기가 없는 (Stagnant)	1	2	3	4	5	생기있는 (Lively)
기계적인 (Mechanical)	1	2	3	4	5	유기적인 (Organic)
인공적인 (Artificial)	1	2	3	4	5	생물적인 (Lifelike)
활발하지 않은 (Inert)	1	2	3	4	5	대화식의 (Interactive)
무관심한 (Apathetic)	1	2	3	4	5	반응을 하는 (Responsive)
Likeability						
싫음 (Dislike)	1	2	3	4	5	좋음 (Like)
친해지기 어려운 (Unfriendly)	1	2	3	4	5	친해지기 쉬운 (Friendly)
불친절한 (Unkind)	1	2	3	4	5	친절한 (Kind)
불쾌한 (Unpleasant)	1	2	3	4	5	유쾌한 (Pleasant)
형편없는 (Awful)	1	2	3	4	5	좋은 (Nice)
Perceived Intelligence						
무능한 (Incompetent)	1	2	3	4	5	유능한 (Competent)
무지한 (Ignorant)	1	2	3	4	5	박식한 (Knowledgeable)
무책임한 (Irresponsible)	1	2	3	4	5	책임감 있는 (Responsible)
무식한 (Unintelligent)	1	2	3	4	5	지적인 (Intelligent)
어리석은 (Foolish)	1	2	3	4	5	현명한 (Sensible)

E.2 Social Presence and Perceived Personalization Items

The NMSPI social-presence items [Biocca et al.(2003)] were translated from English and administered in Korean. The perceived-personalization items were adapted from Komiak and Benbasat’s personalization items [Komiak and Benbasat(2000)] the source items used “RA” as the referent, which was adapted to the avatar in the Korean-administered version.

Table 4: NMSPI and perceived-personalization items: English source items and Korean-administered translations/adaptations.

English source item	Korean-administered translation/adaptation
Co-presence	
I noticed the avatar.	나는 아바타를 인식했다.
The avatar noticed me.	아바타는 나를 인식했다.
The avatar’s presence was obvious to me.	아바타의 존재가 나에게 분명하게 느껴졌다.
My presence was obvious to the avatar.	나의 존재가 아바타에게 분명하게 느껴졌다.
The avatar caught my attention.	아바타는 나의 주의를 끌었다.
I caught the avatar’s attention.	나는 아바타의 주의를 끌었다.
Attentional allocation	
I was easily distracted from the avatar when other things were going on. (R)	다른 일이 일어났을 때, 나는 아바타로부터 쉽게 주의가 분산되었다. (R)
The avatar was easily distracted from me when other things were going on. (R)	다른 일이 일어났을 때, 아바타는 나로부터 쉽게 주의가 분산되었다. (R)
I remained focused on the avatar throughout our interaction.	상호작용 내내 나는 아바타에게 집중했다.
The avatar remained focused on me throughout our interaction.	상호작용 내내 아바타는 나에게 집중했다.
The avatar did not receive my full attention. (R)	아바타는 나에게 충분한 주의를 기울이지 않았다. (R)
I did not receive the avatar’s full attention. (R)	나는 아바타에게 충분한 주의를 기울이지 않았다. (R)
Message understanding	
My thoughts were clear to the avatar.	나의 생각은 아바타에게 명확하게 전달되었다.
The avatar’s thoughts were clear to me.	아바타의 생각은 나에게 명확하게 전달되었다.
It was easy to understand the avatar.	아바타를 이해하기 쉬웠다.
The avatar found it easy to understand me.	아바타는 나를 이해하기 쉬웠을 것이다.
Understanding the avatar was difficult. (R)	아바타를 이해하기 어려웠다. (R)
The avatar had difficulty understanding me. (R)	아바타는 나를 이해하는 데 어려움을 겪었다. (R)
Affective understanding	
I could tell how the avatar felt.	나는 아바타가 어떻게 느끼는지 알 수 있었다.
The avatar could tell how I felt.	아바타는 내가 어떻게 느끼는지 알 수 있었다.
The avatar’s emotions were not clear to me. (R)	아바타의 감정은 나에게 분명하게 드러나지 않았다. (R)
My emotions were not clear to the avatar. (R)	나의 감정은 아바타에게 분명하게 드러나지 않았다. (R)
I could describe the avatar’s feelings accurately.	나는 아바타의 감정을 정확하게 묘사할 수 있었다.
The avatar could describe my feelings accurately.	아바타는 나의 감정을 정확하게 묘사할 수 있었다.
Affective interdependence	
I was sometimes influenced by the avatar’s moods.	나는 때때로 아바타의 기분에 영향을 받았다.
The avatar was sometimes influenced by my moods.	아바타는 때때로 나의 기분에 영향을 받았다.
The avatar’s feelings influenced the mood of our interaction.	아바타의 감정은 우리 상호작용의 분위기에 영향을 미쳤다.
My feelings influenced the mood of our interaction.	나의 감정은 우리 상호작용의 분위기에 영향을 미쳤다.
The avatar’s attitudes influenced how I felt.	아바타의 태도는 내가 느끼는 감정에 영향을 미쳤다.
My attitudes influenced how the avatar felt.	나의 태도는 아바타가 느끼는 감정에 영향을 미쳤다.
Behavioral interdependence	
My behavior was often in direct response to the avatar’s behavior.	나의 행동은 종종 아바타의 행동에 대한 직접적인 반응이었다.
The behavior of the avatar was often in direct response to my behavior.	아바타의 행동은 종종 나의 행동에 대한 직접적인 반응이었다.
I reciprocated the avatar’s actions.	나는 아바타의 행동에 호응하여 반응했다.
The avatar reciprocated my actions.	아바타는 나의 행동에 호응하여 반응했다.
The avatar’s behavior was closely tied to my behavior.	아바타의 행동은 나의 행동과 밀접하게 연결되어 있었다.
My behavior was closely tied to the avatar’s behavior.	나의 행동은 아바타의 행동과 밀접하게 연결되어 있었다.
Perceived personalization	
This avatar understands my needs.	아바타는 나의 취향과 선호를 파악하고 있는 것 같았다.
This avatar knows what I want.	아바타는 내가 어떤 사람인지 알고 있는 것 같았다.
This avatar takes my needs as its own preferences.	아바타의 응답은 나에게 맞춰져 있다고 느꼈다.

For NMSPI, items marked “R” were reverse-coded for analysis. For perceived personalization, the source referent was adapted from “RA” to the avatar in the Korean-administered version.

E.3 Study-Specific Satisfaction, Latency, and Memory Items

The following study-specific items were written in Korean and administered as part of the post-condition or post-study questionnaires. English translations are provided for reporting.

Table 5: Study-specific questionnaire items: Korean-administered items and English translations.

Construct	Korean-administered item	English translation
Overall satisfaction	전반적으로, 이 아바타와의 상호작용에 만족했다.	Overall, I was satisfied with this avatar interaction.
Perceived latency	대화 중 이 아바타의 응답 속도는 전반적으로 어떠했습니까?	Overall, how fast was this avatar’s response during the conversation?
Perceived memory	이 아바타는 나에 대한 정보를 알고 있는 것처럼 행동했다.	This avatar acted as if it knew information about me.
Perceived memory	이 아바타는 나에 대한 정보를 대화에서 적절히 활용했다.	This avatar appropriately used information about me in the conversation.
Perceived memory	이 아바타는 처음 만난 사이가 아닌 것처럼 느껴졌다.	This avatar felt as if we were not meeting for the first time.

The perceived-latency item used a 7-point scale anchored 1 = very fast (매우 빨랐다), 4 = appropriate (적절했다), and 7 = very slow (매우 느렸다). The other study-specific items used 5-point Likert scales. Perceived memory was administered at the end of the session for each condition.

E.4 Open-ended Interview Question

After each condition, participants answered one open-ended question about the conversation they had just completed. After all three conditions, the experimenter conducted a semi-structured interview. The semi-structured interview covered memory, latency, understanding, satisfaction, and salient interaction moments.

Table 6: Post-condition open-ended interview question.

ID	Korean-administered question	English translation
Q1	방금 대화에서 떠오르는 인상이나 느낀 점을 자유롭게 말씀해 주세요. 길이는 상관없고, 떠오르는 게 없으시면 “특별히 없다”도 괜찮습니다.	Please freely share any impressions or feelings that come to mind from the conversation you just had. There is no length requirement; if nothing comes to mind, “nothing in particular” is also fine.

References

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